ADITYA GHOSH

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EDUCATION

Georgia Institute of Technology — Aug 2025 – May 2027
Master of Science in Computer Science — Atlanta, Georgia
Coursework: Machine Learning Systems, Deep Learning in Text Data, Deterministic Optimization — GPA: 3.87

Vellore Institute of Technology — Sep 2021 – Jul 2025
Bachelor of Technology in Computer Science and Engineering — Vellore, India
Specialization: Data Science — GPA: 3.96

EXPERIENCE

TRACTIAN — Atlanta, Georgia
Machine Learning Intern — *May 2026 – Aug 2026*

- Built a CatBoost contract-renewal model using leakage-safe point-in-time features and walk-forward validation across 10+ classifiers, increasing identification of non-renewing customers three months before contract expiry from 34% to 68%.
- Built a RAG system over internal FAQ articles and customer-education video transcripts that returned grounded answers and surfaced relevant video sections for customer questions.
- Developed a Slack bot that notified Technical Account Managers when key usage metrics dropped, operationalizing an interpretable early signal for customer-risk intervention.

Georgia Institute of Technology — Atlanta, Georgia
Graduate Assistant — *Dec 2025 – Aug 2026*

- Developed a cross-modal computer vision pipeline using optical microscopy images with SEM-derived labels to predict nanoscale additive-manufacturing quality, achieving 97.8% grouped accuracy for structure-level classification.
- Built area-level classification models to identify localized under- and over-curing across print regions, achieving 86.2% average accuracy and up to 93.1% accuracy on individual region classes. Paper submitted to *Additive Manufacturing*.

Feola Lab, Emory University — Atlanta, Georgia
Student Researcher — *Jan 2026 – May 2026*

- Engineered a multi-modal 9-class ophthalmic image segmentation model using a ResNet50 backbone across fundus and OCT datasets, implementing weighted cross-entropy and soft Dice loss, achieving 0.81–0.99 Dice scores.
- Applied warm-start transfer learning for cross-dataset fine-tuning and managed experiment tracking, hyperparameter tuning, and performance logging using Weights & Biases.

Indian Statistical Institute — Kolkata, India
Research Intern — *Jun 2024 – Jul 2024*

- Engineered a low-resource training pipeline by fine-tuning a 4-bit quantized Mistral-7B model with LoRA adapters, reducing training time by 50% while preserving model quality and reducing memory usage.
- Expanded query understanding by converting short queries into detailed questions and evaluated semantic similarity using SBERT embeddings, achieving 0.7218 average cosine similarity.

PROJECTS

vLLM Speculative Decoding Cost Modeling

- Benchmarked and instrumented DSpark adaptive verification in vLLM on H100 GPUs with FlashAttention-3, identifying context-length and CUDA Graph effects causing up to 203% step-time underprediction.
- Validated a cache-aware correction that reduced median held-out prediction error from 33.9% to 0.8%.

Natural Language Inference–Based Hallucination Detection (Data4Good 2025)

- Won 1st place in the East region out of 1,015 teams in the national Data4Good Case Challenge with 96.5% final accuracy.
- Built an ensemble of a fine-tuned DeBERTa-MNLI classifier and Azure OpenAI LLM-as-a-judge to detect factual consistency, contradiction, and irrelevance in generated responses.

Distributed Edge AI Voice Assistant

- Built a self-hosted LLM serving stack across heterogeneous GPUs, deploying quantized Qwen models on GTX 1650 and RTX 4060 nodes with Kubernetes-based orchestration and health-aware inference routing.
- Developed a resource-aware scheduler that dynamically routes inference based on node and model availability, scaling fallback models on demand to reclaim 3.4 GiB VRAM and 1 GiB RAM when idle.
- Built GPU observability with Prometheus, Grafana, OpenTelemetry, and a custom NVML exporter exposing per-workload utilization, VRAM, power, temperature, and health metrics while reducing monitoring memory from 405 MiB to 37 MiB.

TECHNICAL SKILLS

Languages	Python, SQL, C++, R
ML/DL	PyTorch, TensorFlow, Keras, Scikit-learn, CatBoost, ResNet, Transformers, LoRA
GenAI	RAG, Embeddings, LangChain, LangGraph, Weaviate, Azure OpenAI
Deployment	AWS, GCP, REST APIs, PostgreSQL, MongoDB, W&B, OpenTelemetry, Prometheus, Grafana
Systems	vLLM, CUDA, FlashAttention, Docker, Kubernetes